Lecture 4.2

Rational Numbers and Direct Proofs

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Birzeit University · Number Theory and Methods of Proof

Why study rational numbers?

A mathematical definition is a tool for proving new statements.

How do we show that a sum of arbitrary rational numbers is rational?

We must construct a ratio of integers with a nonzero denominator.

Today: recognize → formalize → prove → reuse established theorems.

Learning goals

1. Distinguish rational, irrational, and undefined expressions.
2. Represent integers and repeating decimals as rational numbers.
3. Construct a proof using arbitrary rational numbers.
4. Prove or disprove closure claims, with the correct conditions.
5. Derive corollaries and apply established parity properties.

Definition — Rational numbers

$$r \in \mathbb{Q} \iff r = a/b, \text{ where } a,b \in \mathbb{Z} \text{ and } b \neq 0$$

Numerator: any integer (including 0).

Denominator: an integer that MUST be nonzero.

The fraction does not need to be in lowest terms.

Classify: rational, irrational, undefined

RATIONAL

$10/3$, $-5/39$, 0.281, 7, 0, and $0.121212\ldots$ (repeating).
All admit integer ratios.

IRRATIONAL

$\sqrt{2}$ is real but is not a quotient of two integers.
An irrational number must be real.

UNDEFINED

$2/0$ is not a number.
It is neither rational nor irrational.
The denominator condition matters.

Finite decimals and integers are rational

Finite decimal

$$0.281 = 281/1000$$

A terminating decimal can be written with a power of 10 as denominator.

Every integer is rational

$$n = n/1 \quad (n \in \mathbb{Z})$$

Examples: $-3 = -3/1$; $0 = 0/1$. This proves Theorem 4.2.1 for any integer n.

Repeating decimal → rational fraction

Let $x = 0.121212\ldots$

Multiply by 100: $100x = 12.121212\ldots$

Subtract: $99x = 12$

Therefore: $x = 12/99 = 4/33 \in \mathbb{Q}$

The repeated block determines a power of ten for cancellation.

Check the boundary case

Is 2/0 an irrational number?

No. Division by zero is undefined; irrational numbers are real numbers.

Key distinction: “not rational” does NOT automatically mean “irrational.”

An expression involving integers

Let $m,n \in \mathbb{Z}$, where $m \neq 0$ and $n \neq 0$.

Is $(m+n)/(mn)$ rational?

$m+n$ and mn are integers (closure of $\mathbb{Z}$).

$mn \neq 0$ by the zero-product property.

So $(m+n)/(mn)$ is rational by definition.

Theorem 4.2.2 — Closure under addition

If $r, s \in \mathbb{Q}$, then $r+s \in \mathbb{Q}$.

To prove a universal statement, take arbitrary r and s.

Unpack the definition of rational for BOTH numbers.

Write their sum as a single fraction and verify every condition.

Proof: sum of two rational numbers (I)

Suppose r and s are arbitrary rational numbers.

By definition, $r = a/b$ and $s = c/d$

for integers a,b,c,d with $b \neq 0$ and $d \neq 0$.

Then $r+s = a/b + c/d = (ad+bc)/(bd)$.

Proof: sum of two rational numbers (II)

The numerator ad+bc is an integer.

The denominator bd is an integer.

Because b ≠ 0 and d ≠ 0, their product bd ≠ 0.

Thus r+s has the form integer / nonzero integer.

∴ r+s ∈ ℚ. Q.E.D.

Direct-proof checklist

START

Choose arbitrary elements satisfying the hypothesis.

UNPACK

Apply definitions. Introduce integer witnesses and nonzero conditions.

CONSTRUCT

Transform the target into the required defining form.

VERIFY

Check every condition; state the conclusion.

More closure properties

Addition	$r+s$	Always
Subtraction	$r-s$	Always
Multiplication	rs	Always
Division	r/s	Only if $s \neq 0$

Guided proof — Product of rationals

Given $r=a/b$ and $s=c/d$, with $a,b,c,d \in \mathbb{Z}$ and $b,d \neq 0$.

$rs = (ac)/(bd)$.

ac and bd are integers; $bd \neq 0$.

Hence $rs \in \mathbb{Q}$.

Ask: Which fact ensures that bd cannot be zero?

Counterexample — An unrestricted quotient

False claim: The quotient of any two rational numbers is rational.

Take $r=1$ and $s=0$. Both are rational.

But $r/s = 1/0$ is undefined.

Correct claim: r/s is rational when $r,s \in \mathbb{Q}$ and $s \neq 0$.

Deriving new results from known properties

After proving theorems, we may reuse them as established facts.

Even + even = even; odd + odd = even.

Even + odd = odd; odd × odd = odd.

An argument must cite the property used at each step.

Parity example — Derive integrality

Suppose a is even and b is odd. Show $(a^2+b^2+1)/2 \in \mathbb{Z}$.

a^2 is even, while b^2 is odd.

So a^2+b^2 is odd; adding 1 makes it even.

Thus $a^2+b^2+1 = 2k$ for some $k \in \mathbb{Z}$.

Therefore $(a^2+b^2+1)/2 = k \in \mathbb{Z}$.

Corollary 4.2.3 — Double of a rational

If $r \in \mathbb{Q}$, then $2r \in \mathbb{Q}$.

Proof: take an arbitrary rational number r.

2r = r+r is the sum of two rational numbers.

By Theorem 4.2.2, $2r \in \mathbb{Q}$.

Application — Two mechanics

Mechanic A finishes one job in 6 h; B in 8 h.

Together for 2 h: rate = 1/6 + 1/8 = 7/24 job/h.

Completed work = 2 × 7/24 = 14/24.

Remaining work = 1 − 14/24 = 10/24.

B finishes alone in (10/24)/(1/8) = 10/3 h.

Computing connection — Exact versus represented values

Mathematically, 1/3 is an exact rational number.

A floating-point variable generally stores only an approximation.

Integer division in many programming languages can truncate.

Distinguish proof about $\mathbb{Q}$ from operations on machine representations.

Concept check — Can you repair the proof?

"Take r=1/4 and s=1/2. Their sum is 3/4, rational. Therefore the sum of ANY two rationals is rational."

Diagnosis: two examples cannot prove a universal claim.

Repair: choose arbitrary r=a/b and s=c/d, then verify (ad+bc)/(bd).

Appendix A — Core concepts

Rational: $r=a/b$ for integers a,b and $b \neq 0$.

Irrational: real but not rational; undefined $\neq$ irrational.

$\mathbb{Z} \subseteq \mathbb{Q}$; rational sums, differences and products are rational.

Division requires a NONZERO rational divisor.

A proof must verify conditions, not merely simplify symbols.

Appendix B — Conceptual connections

Definition → witness integers → algebra → validity conditions.

Universal claim → arbitrary elements; false claim → counterexample.

Theorem 4.2.2 (sum closure) → 2r=r+r (corollary).

Even/odd properties → parity proof → integer quotient.

Mathematical exactness ≠ floating-point representation.

Appendix C — Worked cases

VALID

$0.281 = 281/1000 \in \mathbb{Q}$
$0 = 0/1 \in \mathbb{Q}$
$0.1212\ldots = 4/33 \in \mathbb{Q}$

INVALID

$2/0$ is undefined (neither rational nor irrational). Universal quotient closure fails at divisor zero.

CONSTRUCTED

$r = a/b$, $s = c/d$; $rs = ac/bd \in \mathbb{Q}$, since $ac, bd \in \mathbb{Z}$ and $bd \neq 0$.

Appendix D — Knowledge check (1/2)

Q1 Classify: 0, √2, 2/0, 0.625. Explain.

Q2 Prove that every integer is rational.

Q3 Is every quotient of two rationals rational? Repair the claim.

Q4 Prove that the negative of any rational number is rational.

Q5 Explain the error in a proof using only r=1/4 and s=1/2.

Appendix D — Knowledge check (2/2)

Q6 Prove that the square of a rational number is rational.

Q7 Derive that 3r is rational using established addition closure.

Q8 True/False: 2/0 is irrational because it is not rational.

Q9 If $r,s \in \mathbb{Q}$ and $r < s$, show $(r+s)/2 \in \mathbb{Q}$.

Q10 If a even and b odd, prove $(a^2+b^2+1)/2$ is integer.

Appendix E — Answer key (1/2)

A1 $0, 0.625$ rational; $\sqrt{2}$ irrational; $2/0$ undefined.

A2 For $n \in \mathbb{Z}$, $n = n/1$ with $1 \neq 0$.

A3 False: $1/0$. Correct if the divisor is nonzero.

A4 If $r = a/b$, then $-r = (-a)/b$; $-a \in \mathbb{Z}$ and $b \neq 0$.

A5 Examples do not justify all possible r, s.

Appendix E — Answer key (2/2)

A6 $r^2 = a^2/b^2$; integers and $b^2 \neq 0$.

A7 $2r = r + r$ rational; $3r = 2r + r$ rational.

A8 False: $2/0$ is undefined, not a real number; hence not irrational.

A9 $r + s$ rational, 2 rational and nonzero; quotient rational.

A10 a^2 even, b^2 odd; $a^2 + b^2 + 1$ even, so half integer.

Appendix F — Further insights

1. Density of rationals: if r<s are rational, (r+s)/2 lies between them.

2. Rational closure under finite sums follows by repeatedly applying addition closure.

3. Exact mathematical fractions and stored binary floating-point numbers differ.

References & next lecture

Original course materials — Prof. Mustafa Jarrar

https://www.jarrar.info/courses/DMath/

Updated course resources — COMP233 GitHub

https://github.com/RuwaYafa/BZU_Courses/tree/main/COMP233

Textbook: Susanna S. Epp, Discrete Mathematics with Applications, 4th ed., §4.2.

Source materials also circulated with minor revisions by Dr. Radi Jarrar.

Next: Lecture 4.3 — Divisibility